Exploring alternative formulations of Smoothed Particle Hydrodynamics for complex fluids and solids

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Introduction

Complex fluids and solids can be described by the Symmetric Hyperbolic Thermodynamically Compatible (SHTC) equations. These equations have a Hamiltonian structure and are advantageous in problems with large deformations and shock waves. One of the state variables in the SHTC equations is the distortion field (close to the inverse of the deformation gradient). Recently, a new numerical method solving the SHTC equations was developed based on the Smoothed Particle hydrodynamics (SPH) [3].

In this paper we will explore two different discretization procedures of the distortion field $\mathbf{A}$ inspired by [1]. We will define those discretizations as a weighted least squares projection of $\mathbf{A}$ onto $\mathrm{GL}(n)$ and weighted least squares projection of velocity gradient $\mathbf{L}$ onto $\mathfrak{gl}(n)$.

Both of these methods are closely related, and we will see that for affine distortion field they will coincide. However, in practice they show a very different numericall behaviour.

Firstly we will work with the discretization of $\mathbf{A}$ defined as a least squares projection onto $\mathrm{GL}(n)$. We will show, however that this discretization does not lead to numerically stable approximation of the flow generated by $\dot{\mathbf{A}} = -\mathbf{A}\mathbf{L}$.

Secondly, we will discretize the velocity gradient $\mathbf{L}$ and defined it as a projection onto $\mathfrak{gl}(n)$. We will show, that this discretization does indeed lead to a numerically stable approximation of the flow generated by $\dot{\mathbf{A}} = -\mathbf{A}\mathbf{L}$.

We will derive an SPH discretization for both cases as well as for the definition in [1].

Furthermore we will discuss that for a suitable choice of the weights in the least squares problem we recover exactly the method from [3] and therefore both our method and SHTC-SPH can be viewed as MLS problem obtained with different weights.

We will implement, compare and verify these results numerically on a classical beryllium benchmark.

At the end we will discuss our findings and further research directions.

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Notation

Symbol	Description
$\mathbf{x}$	Eulerian position
$\mathbf{x}_a$	Eulerian position of particle a
$\mathbf{X}_a$	Lagrangian position of particle a
$\mathbf{X}$	Lagrangian position
$\mathbf{A}(\mathbf{x})$	Distortion field at $\mathbf{x}$
$\mathbf{A}_a$	Approximation of Distortion field at particle a
$\mathbf{L}(\mathbf{x})$	Velocity gradient $\nabla \mathbf{v}$ at $\mathbf{x}$
$\mathbf{L}_a$	Approximation of Velocity gradient $\nabla \mathbf{v}$ at $\mathbf{x}_a$
m_a	Mass of particle a
V_a	Volume of particle a
ρ_a	Density at particle a
τ	Relaxation time τ
ϵ_a	Specific inner energy at $\mathbf{x}_a$
ϵ_0 and ϵ_s	The bulk and shear part of the inner energy st. $\epsilon = \epsilon_0 + \epsilon_s$
U	Total energy of the particle system
c_s	Shear sound speed
c_l	Longitudinal sound speed
c_0	Total sound speed
$\mathbf{f}$	Force
$\mathbf{P}_a$ and $\mathbf{Q}_a$	Matrices such that $\mathbf{A}_a = \mathbf{P}_a \mathbf{Q}_a^{-1}$
$\mathbf{S}_a$	$\mathbf{A}_a^T \partial_{\mathbf{A}} \epsilon_a \mathbf{Q}_a^{-1}$
$(\mathbf{x})_i$	The i-th element of $\mathbf{x}$
$(\mathbf{A})_{ij}$	The ij-th element of $\mathbf{A}$
$\mathbf{x} \otimes \mathbf{y}$	Tensor product of $\mathbf{x}$ and $\mathbf{y}$ ie $(\mathbf{x} \otimes \mathbf{y})_{ij} = (\mathbf{x})_i (\mathbf{y})_j$
$\partial_\rho q$	Partial derivative of q wrt ρ
$\partial_{\mathbf{A}} q$	Partial derivative of q wrt $\mathbf{A}$
$\dot{q}$	Time derivative of q
$O(h)$	Error of order h ie $\exists C > 0$ st $\lim_{h \to 0} O(h)/h = C$
$\mathbf{x}_{ab}$	$\mathbf{x}_a - \mathbf{x}_b$
r_{ab} or $\lvert\mathbf{x}_{ab}\rvert$	Absolute value of $\mathbf{x}_a - \mathbf{x}_b$
w_{ab}	SPH kernel at $\lvert\mathbf{x}_a - \mathbf{x}_b\rvert$
∇w_{ab}	Gradient of SPH kernel $w'_{ab}/r_{ab} \mathbf{x}_{ab}$
h	Smoothing length h
$\lVert\mathbf{x}\rVert$	Euclidian norm of $\mathbf{x}$
$\lVert\mathbf{A}\rVert$	Frobenius norm of $\mathbf{A}$
$\sum_a$	sum over all particles in the simulation domain
$\mathrm{GL}(n)$	General linear group $\mathrm{GL}(n)$ ie the set of $n \times n$ invertible matrices
$\mathfrak{gl}(n)$	The Lie algebra $\mathfrak{gl}(n)$ of $n \times n$ real matrices
SHTC equations	Symmetric Hyperbolic Thermodynamicaly Compatible equations
SPH	Smoothed Particle Hydrodynamics
LS	Least Squares
MLS	Moving Least Squares
ODE	Ordinary Differential Equation
st	such that
wrt	with respect to
ie	that is (id est)
eg	for example (exempli gratia)

Methods

The distortion field $\mathbf{A} \in \mathbb{R}^{d\times d}$ plays a prominent role in the SHTC equations. In the perfectly elastic case it is equal to the inverse of the deformation gradient. If we do not assume perfect elasticity the evolution of $\mathbf{A}$ is being governed by:

$$\dot{\mathbf{A}} = -\mathbf{A}\mathbf{L} + \frac{1}{\theta(\tau,\rho)} \tag{1}$$

where $\theta(\tau,\rho)$ is a relaxation function of relaxation time τ and density ρ. It roughly speaking says how much does the material flow. For large τ the material becomes elastic and for $\tau \to 0+$ the material behaves more and more like a Navier—Stokes—Fourier fluid.

In this work we will turn our attention on the case $\tau \to +\infty$ — a perfectly elastic material and the equation of interest will therefore be:

$$\dot{\mathbf{A}} = -\mathbf{A}\mathbf{L} \tag{2}$$

The reason is that once we can solve the equations for a perfectly elastic solid adding dissipation ($\tau < +\infty$) is straightforward. It can be done either by integrating 1 or by relaxing the initial configuration. Reader can follow [3] if he or she wishes to do that.

Distortion as a minimization problem

In this section inspired by we will define the distortion field as a minimizer of the following minimization problem:

$$\mathbf{A}(x) = \underset{\mathbf{A}\in\mathbb{R}^{d\times d}}{\operatorname{argmin}} \sum_b w(\mathbf{x}-\mathbf{x}_b)\|\mathbf{A}(\mathbf{x}-\mathbf{x}_b) - (\mathbf{X}-\mathbf{X}_b)\|^2 \tag{3}$$

The term

$$e_b(\mathbf{x}) := \|\mathbf{A}(\mathbf{x}-\mathbf{x}_b) - (\mathbf{X}-\mathbf{X}_b)\| \tag{4}$$

plays the role of the error by which we fail to approximate the linearized relation:

$$\mathbf{A}(x)d\mathbf{x} - d\mathbf{X} = 0 \tag{5}$$

The equation 63 can be than written compactly as:

$$\mathbf{A}(x) = \underset{\mathbf{A}\in\mathbb{R}^{d\times d}}{\operatorname{argmin}} \sum_b w(\mathbf{x}-\mathbf{x}_b)e_b(\mathbf{x})^2 \tag{6}$$

which is a weighted least squares problem.

Evaluated at position $\mathbf{x}_a$ associated with particle a we get the following:

$$\mathbf{A}_a = \underset{\mathbf{A}\in\mathbb{R}^{d\times d}}{\operatorname{argmin}} \sum_b w_{ab}\|\mathbf{A}\mathbf{x}_{ab} - \mathbf{X}_{ab}\|^2 \tag{7}$$

Next we will prove that 63 has a solution:

Lemma: (LS problem) 1 *The LS problem 63 has a solution.*

Proof:
Let $I(\mathbf{A}_a) := \frac{1}{2}\sum_b w_{ab}|\ \mathbf{A}_a\mathbf{x}_{ab} - \mathbf{X}_{ab}\|^2$ and observe that $I(\mathbf{A}_a)$ is a differentiable and coercive function meaning:

$$I(\mathbf{A}_a)/\|\mathbf{A}_a\| \to +\infty \tag{8}$$

This implies existence of global minimum.

We can also compute this minimum explicitely:
Lets denote $\mathbf{A}_a\mathbf{x}_{ab} - \mathbf{X}_{ab} =: e_{ab}$ and lets take the Gatteaux derivative of $I(\mathbf{A}_a)$ in the direction $\mathrm{d}\mathbf{A}_a$

$$\begin{aligned}
\mathrm{D}I(\mathbf{A}_a)[\mathrm{d}\mathbf{A}_a] := \lim_{\lambda\to 0+}\frac{I(\mathbf{A}_a+\lambda\mathrm{d}\mathbf{A}_a)-I(\mathbf{A}_a)}{\lambda} = \\
\lim_{\lambda\to 0+}\sum_b w_{ab}\left[(e_{ab}+\lambda\mathrm{d}\mathbf{A}_a\mathbf{x}_{ab}, e_{ab}+\lambda\mathrm{d}\mathbf{A}_a\mathbf{x}_{ab}) - (e_{ab},e_{ab})\right] = \\
\lim_{\lambda\to 0+}\frac{1}{2\lambda}\sum_b w_{ab}\left(2(e_{ab},\lambda\mathrm{d}\mathbf{A}_a\mathbf{x}_{ab}) + \lambda^2\left(\mathrm{d}\mathbf{A}_a\mathbf{x}_a b, \mathrm{d}\mathbf{A}_a\mathbf{x}_a b\right)\right. = \\
\sum_b w_{ab}\left(\mathbf{A}_a\mathbf{x}_{ab}-\mathbf{X}_{ab}\right)\cdot\mathrm{d}\mathbf{A}_a\mathbf{x}_{ab} = \\
\mathrm{d}\mathbf{A}_a : \left(\sum_b w_{ab}\left(\mathbf{A}_a\mathbf{x}_{ab}-\mathbf{X}_{ab}\right)\cdot\mathbf{x}_{ab}\right)
\end{aligned} \tag{9}$$

Which immediately implies:

$$\mathbf{A}_a = \mathbf{P}_a\mathbf{Q}_a^{-1} \tag{10}$$

where

$$\mathbf{P}_a = \left(\sum_b w_{ab}\mathbf{X}_{ab}\otimes\mathbf{x}_{ab}\right) \text{ and } \mathbf{Q}_a = \left(\sum_b w_{ab}\mathbf{x}_{ab}\otimes\mathbf{x}_{ab}\right) \tag{11}$$

Notice that the matrix $\mathbf{Q}_a$ is positive definite and hence invertible (unless $\mathbf{x}_{ab} = 0$ for all a, b). To prove it, we need to show that $\mathbf{v}\cdot\mathbf{Q}_a\mathbf{v} > 0$ for each nonzero $\mathbf{v}$.

$$\mathbf{v}\cdot\mathbf{Q}_a\mathbf{v} = \mathbf{v}\cdot\left(\sum_b w_{ab}\mathbf{x}_{ab}\otimes\mathbf{x}_{ab}\right)\mathbf{v} = \sum_b w_{ab}(\mathbf{x}_{ab}\cdot\mathbf{v})^2 > 0 \tag{12}$$

We can also generalize this and use the definition from Pavelka & Hutter [1] where we hope to achieve better precision at the cost of summing over all particle pairs instead of singletons.

$$\mathbf{A}^{double}(x) = \underset{\mathbf{A}\in\mathbb{R}^{d\times d}}{\operatorname{argmin}}\sum_b\sum_c w(\mathbf{x}-\mathbf{x}_b)w(\mathbf{x}-\mathbf{x}_c)\|\mathbf{A}\mathbf{x}_{bc}-\mathbf{X}_{bc}\|^2 \tag{13}$$

Evaluated at position $\mathbf{x}_a$ associated with particle a we get the following:

$$\mathbf{A}_a^{double} = \underset{\mathbf{A}\in\mathbb{R}^{d\times d}}{\operatorname{argmin}}\sum_b\sum_c w_{ab}w_{ac}\|\mathbf{A}\mathbf{x}_{bc}-\mathbf{X}_{bc}\|^2 \tag{14}$$

Following the same procedure as in the first example we get the explicit formula for $\mathbf{A}^{double}$ in this case:

$$\mathbf{A}^{double}(\mathbf{x}) = \left(\sum_{a,b} w_a w_b\mathbf{X}_{ab}\otimes\mathbf{x}_{ab}\right)\left(\sum_{a,b} w_a w_b\mathbf{x}_{ab}\otimes\mathbf{x}_{ab}\right)^{-1} \tag{15}$$

Evaluated at position $\mathbf{x}_a$ associated with particle a we get the following:

$$\mathbf{A}_a^{double} = \mathbf{P}_a^{double}(\mathbf{Q}_a^{double})^{-1} \tag{16}$$

where

$$\mathbf{P}_a^{double} = \left(\sum_{b,c} w_{ab}w_{ac}\mathbf{X}_{bc}\otimes\mathbf{x}_{bc}\right) \text{ and } \mathbf{Q}_a^{double} = \left(\sum_{b,c} w_{ab}w_{ac}\mathbf{x}_{bc}\otimes\mathbf{x}_{bc}\right) \tag{17}$$

We can make the following interesting observation that allows us to rewrite $\mathbf{A}^d ouble_a$ in an alternative way:

Notice:

$$\mathbf{x}_{bc}\otimes\mathbf{x}_{bc} = (\mathbf{x}_b-\mathbf{x}_c)\otimes(\mathbf{x}_b-\mathbf{x}_c) = \mathbf{x}_b\otimes\mathbf{x}_b + \mathbf{x}_c\otimes\mathbf{x}_c - \mathbf{x}_b\otimes\mathbf{x}_c - \mathbf{x}_c\otimes\mathbf{x}_b \tag{18}$$

Therefore we have:

$$\begin{aligned}\mathbf{Q}_a^{double} &= \sum_{b,c} w_{ab} w_{ac} \left(\mathbf{x}_b \otimes \mathbf{x}_b + \mathbf{x}_c \otimes \mathbf{x}_c - \mathbf{x}_b \otimes \mathbf{x}_c - \mathbf{x}_c \otimes \mathbf{x}_b\right) = \\ &2\left(\sum_b w_{ab}\right)\left(\sum_b w_{ab}\mathbf{x}_b \otimes \mathbf{x}_b\right) - 2\left(\sum_b w_{ab}\mathbf{x}_b\right) \otimes \left(\sum_b w_{ab}\mathbf{x}_b\right)\end{aligned} \tag{19}$$

which after introducing the notation: $\bar{\mathbf{x}}_a := (\sum_b w_{ab}\mathbf{x}_b)/(\sum_b w_{ab})$ and realizing that $\sum_b w_{ab}(\mathbf{x}_b - \bar{\mathbf{x}}_a) = 0$ can be written as:

$$\mathbf{Q}_a^{double} = 2\left(\sum_b w_{ab}\right)\left(\sum_b (\mathbf{x}_b - \bar{\mathbf{x}}_a) \otimes (\mathbf{x}_b - \bar{\mathbf{x}}_a)\right) \tag{20}$$

And similarly:

$$\mathbf{P}_a^{double} = 2\left(\sum_b w_{ab}\right)\left(\sum_b (\mathbf{X}_b - \bar{\mathbf{X}}_a) \otimes (\mathbf{x}_b - \bar{\mathbf{x}}_a)\right) \tag{21}$$

We can also get rid of the prefactor $2\left(\sum_b w_{ab}\right)$ as it will not change the value of $\mathbf{A}_a^{double}$ and simply write:

$$\mathbf{A}_a^{double} = \left(\sum_b (\mathbf{X}_b - \bar{\mathbf{X}}_a) \otimes (\mathbf{x}_b - \bar{\mathbf{x}}_a)\right)\left(\sum_b (\mathbf{x}_b - \bar{\mathbf{x}}_a) \otimes (\mathbf{x}_b - \bar{\mathbf{x}}_a)\right)^{-1} \tag{22}$$

Notice that to compute $\mathbf{A}_a^{double}$ from this definition we only need to know $\bar{\mathbf{x}}_a := (\sum_b w_{ab}\mathbf{x}_b)/(\sum_b w_{ab})$ which can be precomputed.

Next we will make an observation about the order of approximation of $\mathbf{A}_a$ and $\mathbf{A}_a^{double}$: Lets start with the following taylor expansion:

$$\mathbf{X}_b = \mathbf{X}_a + \mathbf{A}(\mathbf{x}_a)\mathbf{x}_{ba} + \frac{1}{2}\nabla\mathbf{A}(\mathbf{x}_a)[\mathbf{x}_{ba}, \mathbf{x}_{ba}] + O(h^3) \tag{23}$$

we can plug it in the definition of $\mathbf{P}_a$ to get:

$$\begin{aligned}\mathbf{P}_a &= \sum_b w_{ab}\left(\mathbf{A}(\mathbf{x}_a)\mathbf{x}_{ba} + \frac{1}{2}\nabla\mathbf{A}(\mathbf{x}_a)[\mathbf{x}_{ba}, \mathbf{x}_{ba}] + O(h^3)\right) \otimes \mathbf{x}_{ab} = \\ &\mathbf{A}(\mathbf{x}_a)\left(\sum_b w_{ab}\mathbf{x}_{ab} \otimes \mathbf{x}_{ab} + O(h^3)\right) = \mathbf{A}(\mathbf{x}_a)\mathbf{Q}_a + O(h^3)\end{aligned} \tag{24}$$

And finally from the definition of $\mathbf{A}_a$ and using the fact that $\mathbf{Q}_a^{-1} = O(h^2)$ we get:

$$\mathbf{A}_a = \mathbf{P}_a\mathbf{Q}_a^{-1} = \left(\mathbf{A}(\mathbf{x}_a) + O(h^3)\right)\mathbf{Q}_a^{-1} = \mathbf{A}(\mathbf{x}_a) + O(h) \tag{25}$$

For $\mathbf{A}^{double}$ we can proceed similarly (using the notation $\nabla\mathbf{A}^{double}(\mathbf{x}_a) =: \mathbf{H}(\mathbf{x}_a)$):

$$\begin{aligned}\mathbf{X}_{ba} &= \mathbf{A}(\mathbf{x}_a)\mathbf{x}_{ba} + \frac{1}{2}\mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ba}, \mathbf{x}_{ba}] + O(h^3) \\ \mathbf{X}_{ca} &= \mathbf{A}(\mathbf{x}_a)\mathbf{x}_{ca} + \frac{1}{2}\mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ca}, \mathbf{x}_{ca}] + O(h^3)\end{aligned} \tag{26}$$

taking the difference of these two expansions we get:

$$\mathbf{X}_{bc} = \mathbf{A}(\mathbf{x}_a)\mathbf{x}_{bc} + \frac{1}{2}\left(\mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ba}, \mathbf{x}_{ba}] - \mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ca}, \mathbf{x}_{ca}]\right) + O(h^3) \tag{27}$$

pluggin this in the definition of $\mathbf{P}_a^{double}$ and swapping the indecees in the last step we get:

$$\begin{aligned}\mathbf{P}_a^{double} &= \left(\sum_{b,c} w_{ab}w_{ac}\mathbf{A}(\mathbf{x}_a)\mathbf{x}_{bc} \otimes \mathbf{x}_{bc} + w_{ab}w_{ac}\frac{1}{2}\left(\mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ba}, \mathbf{x}_{ba}] - \mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ca}, \mathbf{x}_{ca}]\right) \otimes \mathbf{x}_{bc}\right) + O(h^4) = \\ \mathbf{A}(\mathbf{x}_a)\mathbf{Q}_a^{double} &+ \sum_{b,c} w_{ab}w_{ac}\frac{1}{2}\left(\mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ba}, \mathbf{x}_{ba}] - \mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ca}, \mathbf{x}_{ca}]\right) \otimes \mathbf{x}_{bc} + O(h^4) = \mathbf{A}(\mathbf{x}_a)\mathbf{Q}_a^{double} + O(h^4)\end{aligned} \tag{28}$$

Finally we get:

$$\mathbf{A}^{double} = \mathbf{P}_a^{double}(\mathbf{Q}_a^{double})^{-1} = \mathbf{A}(\mathbf{x}_a) + O(h^2) \tag{29}$$

Particle distortion field

Next we will compute the differential of the distortion field wrt a small perturbation in particle possition. Utilizing the definition of $\mathbf{A}_a$ the differential of the distortion field with respect to position $\mathbf{x}_a$ is then equal to:

$$\mathrm{d}\mathbf{A}_a = \mathrm{d}\mathbf{P}_a\mathbf{Q}_a^{-1} + \mathbf{P}_a\mathrm{d}\mathbf{Q}_a^{-1} = \mathrm{d}\mathbf{P}_a\mathbf{Q}_a^{-1} - \mathbf{P}_a\mathbf{Q}_a^{-1}\mathrm{d}\mathbf{Q}_a\mathbf{Q}_a^{-1} \tag{30}$$

Which under the assumptions from [1]:

$$\dot{\mathbf{x}}_a \approx \dot{\mathbf{x}}_b + \nabla\mathbf{v}\Big|_{\mathbf{x}_a} \cdot \mathbf{x}_{ab} \text{ and } \mathbf{X}_{ab} \approx \mathbf{A}_a\mathbf{x}_{ab} \tag{31}$$

simplifies to:

$$\mathrm{d}\mathbf{A}_a = -\mathbf{A}_a\nabla\mathbf{v}(\mathbf{x})\Big|_{\mathbf{x}_a} \tag{32}$$

giving us the evolution equation for the distortion field which is the evolution equation for the distortion field in the SHTC equations.
Computing $\mathrm{d}\mathbf{A}_a$ directly without any approximations yields:

$$\begin{aligned}\mathrm{d}\mathbf{A}_a = -\mathbf{A}_a\mathbf{L}_a + &\left(\sum_b \nabla w_{ab}\cdot \mathrm{d}\mathbf{x}_{ab}\left(\mathbf{X}_{ab} - \mathbf{A}_a\mathbf{x}_{ab}\right)\otimes\mathbf{x}_{ab}\right)\mathbf{Q}_a^{-1} + \\ &\left(\sum_b w_{ab}\left(\mathbf{X}_{ab} - \mathbf{A}_a\mathbf{x}_{ab}\right)\otimes \mathrm{d}\mathbf{x}_{ab}\right)\mathbf{Q}_a^{-1}\end{aligned} \tag{33}$$

where:

$$\mathbf{L}_a := \left(\sum_b w_{ab}\mathbf{v}_{ab}\otimes\mathbf{x}_{ab}\right)\left(\sum_b w_{ab}\mathbf{x}_{ab}\otimes\mathbf{x}_{ab}\right)^{-1} \tag{34}$$

The explicit formulas for $\mathrm{d}\mathbf{P}_a$ and $\mathrm{d}\mathbf{Q}_a$ are:

$$\begin{aligned}\mathrm{d}\mathbf{P}_a &= \sum_b w_{ab}\mathbf{X}_{ab}\otimes\mathrm{d}\mathbf{x}_{ab} + \sum_b \nabla w_{ab}\cdot\mathrm{d}\mathbf{x}_{ab}\mathbf{X}_{ab}\otimes\mathbf{x}_{ab} \\ \mathrm{d}\mathbf{Q}_a &= \sum_b \nabla w_{ab}\cdot\mathrm{d}\mathbf{x}_{ab}\mathbf{x}_{ab}\otimes\mathbf{x}_{ab} + \sum_b w_{ab}\mathbf{x}_{ab}\otimes\mathrm{d}\mathbf{x}_{ab} + \sum_b w_{ab}\mathrm{d}\mathbf{x}_{ab}\otimes\mathbf{x}_{ab}\end{aligned} \tag{35}$$

Equations of motion for A

In this section we will derive the equations of motion.

Density

To proceed we will need the differential of the SPH density

$$\rho_a = \sum_b w_{ab}m_b \tag{36}$$

A straightforward computation yields:

$$\mathrm{d}\rho_a = \sum_b m_b\nabla w_{ab}\cdot\mathrm{d}\mathbf{x}_{ab} \tag{37}$$

Energy

Let $U = \sum_b \epsilon_b$ be the total energy of the particle system, where ϵ_b is the specific energy density of particle b. We will assume that the specific energy density depends on the density ρ and the distortion field $\mathbf{A}$ ie. $\epsilon_b = \epsilon_b(\rho_b, \mathbf{A}_b)$.

We will also assume that the inner energy ϵ decomposes additively into bulk and shear part as $\epsilon(\rho, \mathbf{A}) = \epsilon_0(\rho) + \epsilon_s(\mathbf{A})$.

We will use the following quadratic relation for the bulk component:

$$\begin{aligned}\epsilon_0(\rho) &= c_0^2/2\left(\frac{\rho_0}{\rho} - 1\right)^2 \\ \epsilon_0'(\rho) &= c_0^2\left(\frac{\rho_0}{\rho} - 1\right)\left(-\rho_0/\rho^2\right) = c_0^2\rho_0\left(\frac{1}{\rho^2} - \frac{\rho_0}{\rho^3}\right)\end{aligned} \tag{38}$$

For the shear component we will use the following equation:

$$\begin{aligned}\epsilon_s(\mathbf{A}) &= \frac{c_s^2}{4}\|\mathrm{dev}(\mathbf{A}^T\mathbf{A})\|_F^2 \\ \epsilon_s'(\mathbf{A}) &= c_s^2\mathbf{A}\mathrm{dev}(\mathbf{A}^T\mathbf{A})\end{aligned} \tag{39}$$

Next we will compute the differential of the energy and split it into the bulk part $\mathrm{d}U_0$ and the shear part $\mathrm{d}U_s$ as:

$$\mathrm{d}U = \sum_a m_a\left(\frac{\partial\epsilon_a}{\partial\rho}\mathrm{d}\rho_a + \frac{\partial\epsilon_a}{\partial\mathbf{A}} : \mathrm{d}\mathbf{A}_a\right) = \mathrm{d}U_0 + \mathrm{d}U_s \tag{40}$$

For the bulk part we get the following:

$$\mathrm{d}U_0 = \sum_a m_a\frac{\partial\epsilon_a}{\partial\rho}\mathrm{d}\rho_a = \sum_{a,b} m_a m_b \partial_\rho\epsilon_a\nabla w_{ab}\cdot(\mathrm{d}\mathbf{x}_a - \mathrm{d}\mathbf{x}_b) = \sum_{a,b} m_a m_b\left(\partial_\rho\epsilon_a + \partial_\rho\epsilon_b\right)\nabla w_{ab}\cdot\mathrm{d}\mathbf{x}_a \tag{41}$$

And for the shear part we will get:

$$\begin{aligned}\mathrm{d}U_s = \sum_a m_a\frac{\partial\epsilon_a}{\partial\mathbf{A}} : \mathrm{d}\mathbf{A}_a = \sum_a m_a\left(\partial_{\mathbf{A}}\epsilon_a : \left(\mathrm{d}\mathbf{P}_a\mathbf{Q}_a^{-1} - \mathbf{P}_a\mathbf{Q}_a^{-1}\mathrm{d}\mathbf{Q}_a\mathbf{Q}_a^{-1}\right)\right) = \\ \sum_a m_a\left(\partial_{\mathbf{A}}\epsilon_a : \mathrm{d}\mathbf{P}_a\mathbf{Q}_a^{-1} - \partial_{\mathbf{A}}\epsilon_a : \mathbf{A}_a\mathrm{d}\mathbf{Q}_a\mathbf{Q}_a^{-1}\right) = \sum_a m_a\left((A) - (B)\right)\end{aligned} \tag{42}$$

The first part labeled (A) yields:

$$\begin{aligned}\sum_a m_a(A) = \sum_a m_a\left[\partial_{\mathbf{A}}\epsilon_a : \mathrm{d}\mathbf{P}_a\mathbf{Q}_a^{-1}\right] = \sum_a m_a\left[\partial_{\mathbf{A}}\epsilon_a\mathbf{Q}_a^{-1} : \mathrm{d}\mathbf{P}_a\right] = \\ \sum_a m_a\left[\mathbf{A}_a^{-T}\mathbf{S}_a : \left(\sum_b w_{ab}\mathbf{X}_{ab}\otimes\mathrm{d}\mathbf{x}_{ab} + \sum_b\nabla w_{ab}\cdot\mathrm{d}\mathbf{x}_{ab}\mathbf{X}_{ab}\otimes\mathbf{x}_{ab}\right)\right] = \\ \sum_a m_a\left[\left(\sum_b w_{ab}\mathbf{S}_a^T\mathbf{A}_a^{-1}\mathbf{X}_{ab}\cdot\mathrm{d}\mathbf{x}_{ab} + \sum_b\nabla w_{ab}\cdot\mathrm{d}\mathbf{x}_{ab}\mathbf{X}_{ab}\cdot\mathbf{A}_a^{-T}\mathbf{S}_a\mathbf{x}_{ab}\right)\right] = \\ \sum_{a,b} m_a\left(w_{ab}\mathbf{S}_a^T\mathbf{A}_a^{-1}\mathbf{X}_{ab} + \nabla w_{ab}\mathbf{A}_a^{-T}\mathbf{S}_a\mathbf{x}_{ab}\right)\cdot(\mathrm{d}\mathbf{x}_a - \mathrm{d}\mathbf{x}_b) = \\ \sum_{a,b}\left[w_{ab}\left(m_a\mathbf{S}_a^T\mathbf{A}_a^{-1} + m_b\mathbf{S}_b^T\mathbf{A}_b^{-1}\right)\mathbf{X}_{ab} + \nabla w_{ab}\mathbf{X}_{ab}\cdot\left(m_a\mathbf{A}_a^{-T}\mathbf{S}_a + m_b\mathbf{A}_b^{-T}\mathbf{S}_b\right)\mathbf{x}_{ab}\right]\cdot\mathrm{d}\mathbf{x}_a\end{aligned} \tag{43}$$

And the second part labeled (B) yields:

$$\begin{aligned}\sum_a m_a(B) = \sum_a m_a\left[\partial_{\mathbf{A}}\epsilon_a : \mathbf{A}_a\mathrm{d}\mathbf{Q}_a\mathbf{Q}_a^{-1}\right] = \sum_a m_a\left[\mathbf{A}_a^T\partial_{\mathbf{A}}\epsilon_a\mathbf{Q}_a^{-1} : \mathrm{d}\mathbf{Q}_a\right] = \sum_a m_a\left[\mathbf{S}_a : \mathrm{d}\mathbf{Q}_a\right] = \\ \sum_a m_a\left[\mathbf{S}_a : \left(\sum_b\nabla w_{ab}\cdot\mathrm{d}\mathbf{x}_{ab}\mathbf{x}_{ab}\otimes\mathbf{x}_{ab} + \sum_b w_{ab}\mathbf{x}_{ab}\otimes\mathrm{d}\mathbf{x}_{ab} + \sum_b w_{ab}\mathrm{d}\mathbf{x}_{ab}\otimes\mathbf{x}_{ab}\right)\right] = \\ \sum_a m_a\left[\left(\sum_b\nabla w_{ab}\cdot\mathrm{d}\mathbf{x}_{ab}\mathbf{x}_{ab}\cdot\mathbf{S}_a\mathbf{x}_{ab} + \sum_b w_{ab}\mathbf{S}_a^T\mathbf{x}_{ab}\cdot\mathrm{d}\mathbf{x}_{ab} + \sum_b w_{ab}\mathrm{d}\mathbf{x}_{ab}\cdot\mathbf{S}_a\mathbf{x}_{ab}\right)\right] = \\ \sum_a m_a\left[\left(\sum_b\nabla w_{ab}\mathbf{x}_{ab}\cdot\mathbf{S}_a\mathbf{x}_{ab} + \sum_b w_{ab}\mathbf{S}_a^T\mathbf{x}_{ab} + \sum_b w_{ab}\cdot\mathbf{S}_a\mathbf{x}_{ab}\right)\cdot(\mathrm{d}\mathbf{x}_a - \mathrm{d}\mathbf{x}_b)\right] = \\ \sum_{a,b}\left[\nabla w_{ab}\mathbf{x}_{ab}\cdot\left(m_a\mathbf{S}_a + m_b\mathbf{S}_b\right)\mathbf{x}_{ab} + w_{ab}\left(m_a\mathbf{S}_a^T + m_b\mathbf{S}_b^T\right)\mathbf{x}_{ab} + w_{ab}\left(m_a\mathbf{S}_a + m_b\mathbf{S}_b\right)\mathbf{x}_{ab}\right]\cdot\mathrm{d}\mathbf{x}_a\end{aligned} \tag{44}$$

Finally, putting these terms back together we get:

$$\begin{aligned}
&\sum_a m_a\left((A)-(B)\right)= \\
&\sum_{a,b}\Bigg(w_{ab}\left[m_a\mathbf{S}_a^T\left(\mathbf{A}_a^{-1}\mathbf{X}_{ab}-\mathbf{x}_{ab}\right)+m_b\mathbf{S}_b^T\left(\mathbf{A}_b^{-1}\mathbf{X}_{ab}-\mathbf{x}_{ab}\right)\right]+ \\
&\sum_{a,b}\nabla w_{ab}\left[\left(\mathbf{A}_a^{-1}\mathbf{X}_{ab}-\mathbf{x}_{ab}\right)\cdot\left(m_a\mathbf{S}_a\mathbf{x}_{ab}\right)+\left(\mathbf{A}_b^{-1}\mathbf{X}_{ab}-\mathbf{x}_{ab}\right)\cdot\left(m_b\mathbf{S}_b\mathbf{x}_{ab}\right)\right]+ \\
&-\sum_{a,b}w_{ab}\left(m_a\mathbf{S}_a+m_b\mathbf{S}_b\right)\mathbf{x}_{ab}\Bigg)\cdot\mathrm{d}\mathbf{x}_a
\end{aligned} \tag{45}$$

We will further employ the notation $e_{ab}=\mathbf{A}_a^{-1}\mathbf{X}_{ab}-\mathbf{x}_{ab}$ and with that the full differential of energy $\mathrm{d}U$ reads:

$$\begin{aligned}
\mathrm{d}U=\sum_{a,b}\Bigg(&m_am_b\left(\partial_\rho\epsilon_a+\partial_\rho\epsilon_b\right)\nabla w_{ab}-\sum_{a,b}w_{ab}\left(m_a\mathbf{S}_a+m_b\mathbf{S}_b\right)\mathbf{x}_{ab}+ \\
&w_{ab}\left[m_a\mathbf{S}_a^Te_{ab}-m_b\mathbf{S}_b^Te_{ba}\right]+\sum_{a,b}\nabla w_{ab}\left[e_{ab}\cdot\left(m_a\mathbf{S}_a\mathbf{x}_{ab}\right)+e_{ba}\cdot\left(m_b\mathbf{S}_b\mathbf{x}_{ab}\right)\right]\Bigg)\cdot\mathrm{d}\mathbf{x}_a
\end{aligned} \tag{46}$$

Using Newton's second law $m\dot{\mathbf{v}}=-\frac{\mathrm{d}U}{\mathrm{d}\mathbf{x}}$ we get the following equations of motion:

$$\begin{aligned}
m_a\dot{\mathbf{v}}_a=-\sum_{a,b}\Bigg(&m_am_b\left(\partial_\rho\epsilon_a+\partial_\rho\epsilon_b\right)\nabla w_{ab}-\sum_{a,b}w_{ab}\left(m_a\mathbf{S}_a+m_b\mathbf{S}_b\right)\mathbf{x}_{ab}+ \\
&w_{ab}\left[m_a\mathbf{S}_a^Te_{ab}-m_b\mathbf{S}_b^Te_{ba}\right]+\sum_{a,b}\nabla w_{ab}\left[e_{ab}\cdot\left(m_a\mathbf{S}_a\mathbf{x}_{ab}\right)+e_{ba}\cdot\left(m_b\mathbf{S}_b\mathbf{x}_{ab}\right)\right]\Bigg)
\end{aligned} \tag{47}$$

Here we are presented with two options, we can either work with the full system, or we can hope that the contribution of $e_{ab}=\mathbf{A}_a^{-1}\mathbf{X}_{ab}-\mathbf{x}_{ab}a$ will be very small (which should be the case) and we can work with significantly simplier system given by:

$$\dot{\mathbf{x}}_a=\mathbf{v}_a \tag{48a}$$

$$\dot{\mathbf{v}}_a=\sum_b m_b\left(w_{ab}\left(\frac{\mathbf{S}_a}{m_b}+\frac{\mathbf{S}_b}{m_a}\right)-\left(\partial_\rho\epsilon_a+\partial_\rho\epsilon_b\right)\nabla w_{ab}\right) \tag{48b}$$

$$\mathbf{A}_a=\left(\sum_b w_{ab}\mathbf{X}_{ab}\otimes\mathbf{x}_{ab}\right)\left(\sum_b w_{ab}\mathbf{x}_{ab}\otimes\mathbf{x}_{ab}\right)^{-1} \tag{48c}$$

where

$$\mathbf{S}_a=c_s^2\mathbf{A}_a^T\mathbf{A}_a\mathrm{dev}(\mathbf{A}_a^T\mathbf{A}_a)\mathbf{Q}_a^{-1}\quad\text{and}\quad\partial_\rho\epsilon_a=c_0^2\rho_0\left(\frac{1}{\rho_a^2}+\frac{\rho_0}{\rho_a^3}\right) \tag{48d}$$

Observation: Conservational Properties

Our method conserves energy, linear and angular momentum. Because the equations of motion can be derived from the following Lagrangian:

$$\mathcal{L}=\sum_b m_b\left(\frac{1}{2}\mathbf{v}_b^2-\epsilon_b\right) \tag{49}$$

the conservation of linear and angular momentum is a consequence of the invariance of the Lagrangian $\mathcal{L}$ to translations and rotations.

Comparison between SPH and MLS

In this section we compare the main conceptual differences between the SHTC-SPH method developed by Kincl et al. MLS methods and between the distortion definition from [1]. Two disting approaches to approximating $\nabla \mathbf{v}$ arise in meshfree methods. Both approaches are equally valid and internally consistent.

SPH:
Lets first review the approach utilized by SPH. Recall that at the heart of SPH lies the idea of approximating differential operators and scalar functions by first convoluting them with the mollifiying kernel w:

$$\begin{gathered} f = f * \delta \approx f * w \\ \nabla f = \nabla f * \delta \approx \nabla f * w = f * \nabla w \end{gathered} \tag{50}$$

and than approximating the resulting integral with numerical quadrature:

$$\begin{aligned} (f * w)(\mathbf{x}_a) &= \int f(\mathbf{y}) w(\mathbf{x}_a - \mathbf{y}) \mathrm{d}\mathbf{y} \approx \sum_b V_b f(\mathbf{x}_b) w_{ab} \\ (f * \nabla w)(\mathbf{x}_a) &= \int f(\mathbf{y}) \nabla w(\mathbf{x}_a - \mathbf{y}) \mathrm{d}\mathbf{y} = \sum_b V_b f(\mathbf{x}_b) \nabla w_{ab} \end{aligned} \tag{51}$$

using the volumes $V_b = \rho_b / m_b$ of the particles as quadrature weights. Similarly for a vector function $\mathbf{f}$ we have:

$$\begin{gathered} \mathbf{f}(\mathbf{x}_a) \approx \sum_b V_b \mathbf{f}(\mathbf{x}_b) w_{ab} \\ \nabla \mathbf{f}(\mathbf{x}_a) \approx \sum_b V_b \mathbf{f}(\mathbf{x}_b) \otimes \nabla w_{ab} \end{gathered} \tag{52}$$

Recall, that our aim is to get discrete SPH form of the following relation:

$$\mathrm{d}\mathbf{A} = -\mathbf{A} \mathrm{d}\mathbf{L} \tag{53}$$

Following the SPH philosophy we can approximate $\mathbf{L} = \nabla \mathbf{v}$ directly as:

$$\mathbf{L} = \nabla \mathbf{v} = \sum_b V_b \mathbf{v}_{ab} \otimes \nabla w_{ab} \tag{54}$$

However this discretization is not first order exact.

To alleviate this issue a correction matrix is introduced. The corrected approximation of the velocity gradient than reads as:

$$\mathbf{L}^{corrected} = \left(\sum_b V_b \mathbf{v}_{ab} \otimes \nabla w_{ab} \right) \left(\sum_b V_b \mathbf{x}_{ab} \otimes \nabla w_{ab} \right)^{-1} \tag{55}$$

notice that this has the same structure as the approximation arising from the MLS approximation - and this is to be expected as the MLS projection of some (tensor) variable is defined as the best affine aprroximation - which is exactly first order exact.

Kincl uses slightly modified version replacing the volumes V_a with m_a masses of respective particles:

$$\mathbf{L} = \left(\sum_b m_b \mathbf{v}_{ab} \otimes \nabla w_{ab} \right) \left(\sum_b m_b \mathbf{x}_{ab} \otimes \nabla w_{ab} \right)^{-1} \tag{56}$$

for a benchmark problem such as Beryllium plate where density is roughly constant $\rho \approx$ const. this is equivallent to the standard SPH definition introduced above. Another advantage of choosing L in such manner is that it interplays nicely with the differential of density $\mathrm{d}\rho$ and gives us remarkably ellegant system of equations to work with:

$$\mathrm{d}\rho_a = \sum_b m_b \nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} \tag{57}$$

Equations of motion follow from the same procedure as in previous sections and we recover rather elegant system:

$$\dot{\mathbf{x}}_a = \mathbf{v}_a \tag{58a}$$

$$\dot{\mathbf{v}}_a = \sum_b m_b \left(\mathbf{T}_a + \mathbf{T}_b\right) \nabla w_{ab} \tag{58b}$$

$$\dot{\mathbf{A}}_a = -\mathbf{A}_a \mathbf{L}_a \tag{58c}$$

where

$$\mathbf{T}_a := -\partial_\rho \epsilon \mathbf{I} + \mathbf{A}_a^T \partial_{\mathbf{A}} \epsilon \left(\sum_b m_b \mathbf{x}_{ab} \otimes \nabla w_{ab}\right)^{-1} \tag{58d}$$

MLS:

The other approach (refered to as MLS - moving least squares) defines the gradient of a function f as the best affine approximation of it in the sense of a least squares problem

$$\nabla f(\mathbf{x}) = \operatorname*{argmin}_{\mathbf{A} \in \mathbb{R}^{d\times d}} \int w(\mathbf{x}-\mathbf{y}) \| f(\mathbf{y}) - f(\mathbf{x}) - \mathbf{A}\cdot(\mathbf{x}-\mathbf{y})\|^2 \mathrm{d}\mathbf{y} \tag{59}$$

which gives:

$$\nabla f(\mathbf{x}) = \left(\int w(\mathbf{x}-\mathbf{y})(f(\mathbf{y})-f(\mathbf{x})) \otimes (\mathbf{y}-\mathbf{x}) \mathrm{d}\mathbf{y}\right) \left(\int w(\mathbf{x}-\mathbf{y})(\mathbf{y}-\mathbf{x}) \otimes (\mathbf{y}-\mathbf{x}) \mathrm{d}\mathbf{y}\right)^{-1} \tag{60}$$

Motivated by MLS approach we will define the continuum equivallent of our minimalization problem.

$$\mathbf{A}(x) = \operatorname*{argmin}_{\mathbf{A} \in \mathbb{R}^{d\times d}} \sum_b w(\mathbf{x}-\mathbf{x}_b) \|\mathbf{A}(\mathbf{x}-\mathbf{x}_b) - (\mathbf{X}-\mathbf{X}_b)\|^2 \tag{61}$$

which reads as:

$$\mathbf{A}(x) = \operatorname*{argmin}_{\mathbf{A} \in \mathbb{R}^{d\times d}} \int w(\mathbf{x}-\mathbf{y}) \|\mathbf{A}(\mathbf{x}-\mathbf{y}) - (\mathbf{X}-\mathbf{Y})\|^2 \mathrm{d}\mathbf{y} \tag{62}$$

and the SPH discretization we will get the following:

$$\mathbf{A}(x) = \operatorname*{argmin}_{\mathbf{A} \in \mathbb{R}^{d\times d}} \sum_b V_b w(\mathbf{x}-\mathbf{x}_b) \|\mathbf{A}(\mathbf{x}-\mathbf{x}_b) - (\mathbf{X}-\mathbf{X}_b)\|^2 \tag{63}$$

We already now the solution of this problem:

$$\mathbf{A}_a = \left(\sum_b \frac{m_b}{\rho_b} w_{ab} \mathbf{X}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b \frac{m_b}{\rho_b} w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1} \tag{64}$$

Using our previous calculations we also immediately see that:

$$\mathrm{d}\mathbf{A}_a = -\mathbf{A}_a \mathrm{d}\mathbf{L}_a \quad \text{where} \quad \mathrm{d}\mathbf{L}_a = \left(\sum_b V_b w_{ab} \mathrm{d}\mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b V_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1} \tag{65}$$

The evolution equations can be derived analogously as in the previous section:

$$\dot{\mathbf{x}}_a = \mathbf{v}_a \tag{66a}$$

$$m_a \dot{\mathbf{v}}_a = \sum_b m_b m_a \left[(\mathbf{S}_a/\rho_b + \mathbf{S}_b/\rho_a) w_{ab} \mathbf{x}_{ab} - \left(\partial_\rho \epsilon_a + \partial_\rho \epsilon_b\right) \nabla w_{ab} \right] \tag{66b}$$

$$\dot{\mathbf{A}}_a = -\mathbf{A}_a \mathbf{L}_a \tag{66c}$$

Pavelka-Hutter:

Lastly we have the least squares particle estimator defined in [1]. Notice that unlike the two previous this definition does not factor in the varying volumes of different particles and is purely geometrical.

This has to be taken in account while discretizing the equations otherwise this method might provide incorrect results near the areas where particle density changes dramatically or where particles are missing - that is near the boundaries.

Otherwise the method is just a special case of MLS and falls fully within that framework.

Evolution equations for $\mathbf{A}^{double}$

In this section we will derive evolution equations for $\mathbf{A}^{double}$ from Pavelka a Hutter [1]:
Recall the distortion field from [1]:

$$\mathbf{A}_a^{double} := \mathbf{P}^{double}\left(\mathbf{Q}^{double}\right)^{-1} = \left(\sum_{b,c} w_{ab}w_{ac}\mathbf{X}_{bc} \otimes \mathbf{x}_{bc}\right)\left(\sum_{b,c} w_{ab}w_{ac}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc}\right)^{-1}$$

Next, analogously to the previous section our goal is to compute the differential of $\mathbf{A}_a$ and than use Newton's equations of motion to get the evolutionary equations. Proceeding with our plan:

$$\mathrm{d}\mathbf{A}_a = \mathrm{d}\left(\mathbf{P}_a^{double}\left(\mathbf{Q}^{double}\right)_a^{-1}\right) = \mathrm{d}\mathbf{P}^{double}\mathbf{Q}^{double} - \mathbf{A}^{double}\mathrm{d}\mathbf{Q}^{double}\left(\mathbf{Q}^{double}\right)^{-1}$$

where $\mathbf{P}_a^{double}$ and $\mathbf{Q}_a^{double}$ are:

$$\mathrm{d}\mathbf{P}_a^{double} = 2\sum_{b,c}\left(\nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab}\right) w_{ab}\mathbf{X}_{bc} \otimes \mathbf{x}_{bc} + \sum_{b,c} w_{ab}w_{ac}\mathbf{X}_{bc} \otimes \mathrm{d}\mathbf{x}_{bc}$$

$$\mathrm{d}\mathbf{Q}_a^{double} = 2\sum_{b,c}\left(\nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab}\right) w_{ac}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc} + \sum_{b,c} w_{ab}w_{ac}\mathbf{x}_{bc} \otimes \mathrm{d}\mathbf{x}_{bc} + \sum_{b,c} w_{ab}w_{ac}\mathrm{d}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc}$$

plugging these in the previous expression for the differential $\mathrm{d}\mathbf{A}^{double}$ we obtain:

$$\begin{aligned}
\mathrm{d}\mathbf{A}_a^{double} &= 2\sum_{b,c}\left(\nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab}\right) w_{ab}\mathbf{X}_{bc} \otimes \mathbf{x}_{bc} + \sum_{b,c} w_{ab}w_{ac}\mathbf{X}_{bc} \otimes \mathrm{d}\mathbf{x}_{bc}\left(\mathbf{Q}^{double}\right)^{-1} - \\
\mathbf{A}_a^{double}&\left(2\sum_{b,c}\left(\nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab}\right) w_{ac}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc} + \sum_{b,c} w_{ab}w_{ac}\mathbf{x}_{bc} \otimes \mathrm{d}\mathbf{x}_{bc} + \sum_{b,c} w_{ab}w_{ac}\mathrm{d}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc}\right) = \\
&2\left(\sum_{b,c}\left(\nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab}\right) w_{ac}\left(\mathbf{X}_{bc} - \mathbf{A}_a^{double}\mathbf{x}_{bc}\right) \otimes \mathbf{x}_{bc}\right)\left(\mathbf{Q}^{double}\right)^{-1} + \\
&\left(\sum_{b,c} w_{ab}w_{ac}\left(\mathbf{X}_{bc} - \mathbf{A}_a^{double}\mathbf{x}_{bc}\right) \otimes \mathrm{d}\mathbf{x}_{bc}\right)\left(\mathbf{Q}^{double}\right)^{-1} + \\
&-\mathbf{A}_a^{double}\left(\sum_{b,c} w_{ab}w_{ac}\mathrm{d}\mathbf{x}_{bc} \otimes \mathbf{x}_{b}c\right)\left(\mathbf{Q}^{double}\right)^{-1}
\end{aligned}$$

Analogously to the previous section, we can identify the error term $e_{abc} := \mathbf{X}_{bc} - \mathbf{A}_a^{double}\mathbf{x}_{bc}$ and work with the 'simplified' differential. The simplified differential of $\mathbf{A}^{double}$ is equal to:

$$\mathrm{d}\mathbf{A}_a^{double} = -\mathbf{A}_a^{double}\left(\sum_{b,c} w_{ab}w_{ac}\mathrm{d}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc}\right)\left(\sum_{b,c} w_{ab}w_{ac}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc}\right)^{-1} \tag{67}$$

And we can once again idenfity $\mathbf{L}_a^{double}$ as:

$$\mathbf{L}_a^{double} := \left(\sum_{b,c} w_{ab}w_{ac}\mathrm{d}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc}\right)\left(\sum_{b,c} w_{ab}w_{ac}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc}\right)^{-1}$$

and write 67 simply as:

$$\mathrm{d}\mathbf{A}_a^{double} = -\mathbf{A}_a^{double}\mathbf{L}_a^{double}$$

To obtain the evolution we need to compute the differential of energy $\mathrm{d}U$ and arrange the terms in the following manner:

$$\mathrm{d}U = \sum_{b,c} \text{something} \cdot \mathrm{d}\mathbf{x}_a$$

We will continue as in the case for $\mathbf{A}$ and split the differential of energy into the bulk part $\mathrm{d}U_0$ and the shear part $\mathrm{d}U_s$ as:

$$\mathrm{d}U = \sum_a m_a \left(\frac{\partial \epsilon_a}{\partial \rho} \mathrm{d}\rho_a + \frac{\partial \epsilon_a}{\partial \mathbf{A}} : \mathrm{d}\mathbf{A}_a^{double} \right) = \mathrm{d}U_0 + \mathrm{d}U_s$$

For the bulk part we get the same expression as for $\mathbf{A}$:

$$\mathrm{d}U_0 = \sum_{a,b} m_a m_b \left(\partial_\rho \epsilon_a + \partial_\rho \epsilon_b \right) \nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_a$$

And for the shear part we will get:

$$\begin{aligned}
\mathrm{d}U_s = \sum_a m_a \frac{\partial \epsilon_a}{\partial \mathbf{A}} : \mathrm{d}\mathbf{A}_a^{double} = \sum_a m_a \left(-\partial_{\mathbf{A}} \epsilon_a : \mathbf{A}_a^{double} \mathrm{d}\mathbf{Q}_a^{double} \left(\mathbf{Q}^{double} \right)^{-1} \right) = \\
-\sum_a m_a \left(\mathbf{A}_a^{double} \right)^T \partial_{\mathbf{A}} \epsilon_a \left(\mathbf{Q}^{double} \right)^{-1} : \left(\sum_{b,c} w_{ab} w_{ac} \mathrm{d}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc} \right) = \\
-\sum_a m_a \mathbf{S}_a^{double} : \left(\sum_{b,c} w_{ab} w_{ac} \mathrm{d}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc} \right) = \\
-\sum_{a,b,c} m_a w_{ab} w_{ac} \mathbf{S}_a^{double} \mathbf{x}_{bc} \cdot \mathrm{d}\mathbf{x}_{bc} = -\sum_{a,b,c} m_a w_{ab} w_{ac} \mathbf{S}_a^{double} \mathbf{x}_{bc} \cdot \mathrm{d}\mathbf{x}_b + \sum_{a,b,c} m_a w_{ab} w_{ac} \mathbf{S}_a^{double} \mathbf{x}_{bc} \cdot \mathrm{d}\mathbf{x}_c = \\
-\sum_{b,a,c} m_b w_{ab} w_{bc} \mathbf{S}_b^{double} \mathbf{x}_{ac} \cdot \mathrm{d}\mathbf{x}_a + \sum_{c,b,a} m_c w_{cb} w_{ac} \mathbf{S}_c^{double} \mathbf{x}_{ba} \cdot \mathrm{d}\mathbf{x}_a = \\
\sum_{a,b,c} \left(m_b w_{ab} w_{bc} \mathbf{S}_b^{double} \mathbf{x}_{ca} + m_c w_{cb} w_{ac} \mathbf{S}_c^{double} \mathbf{x}_{ba} \right) \cdot \mathrm{d}\mathbf{x}_a = 2 \left(\sum_{a,b,c} m_b w_{bc} w_{ab} \mathbf{S}_b^{double} \mathbf{x}_{ca} \right) \cdot \mathrm{d}\mathbf{x}_a = \\
2 \left(\sum_{a,b} m_b w_{ab} \mathbf{S}_b^{double} \left(\sum_c w_{bc} \mathbf{x}_c - \sum_c w_{bc} \mathbf{x}_a \right) \right) \cdot \mathrm{d}\mathbf{x}_a = \left(\sum_{a,b} m_b w_{ab} 2 W_a \mathbf{S}_b^{double} \left(\bar{\mathbf{x}}_b - \mathbf{x}_a \right) \right) \cdot \mathrm{d}\mathbf{x}_a
\end{aligned}$$

Finally, using Newton's second law $m\dot{\mathbf{v}} = -\frac{\mathrm{d}U}{\mathrm{d}\mathbf{x}}$ we will get the two following (and equivalent) equations of motion for $\mathbf{v}_a$:

$$m_a \dot{\mathbf{v}}_a = -\sum_b \left(m_a m_b \left(\partial_\rho \epsilon_a + \partial_\rho \epsilon_b \right) \nabla w_{ab} - m_b w_{ab} 2 W_a \mathbf{S}_b^{double} \left(\mathbf{x}_a - \bar{\mathbf{x}}_b \right) \right)$$

or equivalently:

$$m_a \dot{\mathbf{v}}_a = -\sum_{b,c} \left(m_a m_b \left(\partial_\rho \epsilon_a + \partial_\rho \epsilon_b \right) \nabla w_{ab} - m_b w_{ab} w_{bc} \mathbf{S}_b^{double} \mathbf{x}_{ca} - m_c w_{cb} w_{ac} \mathbf{S}_c^{double} \mathbf{x}_{ba} \right)$$

The full simplified system of equations for $\mathbf{A}^{double}$ reads as:

$$\dot{\mathbf{x}}_a = \mathbf{v}_a \tag{68a}$$

$$\dot{\mathbf{v}}_a = -\sum_b m_b \Big(\left(\partial_\rho \epsilon_a + \partial_\rho \epsilon_b \right) \nabla w_{ab} + \frac{2}{m_b} w_{ab} W_a \mathbf{S}_b^{double} \left(\mathbf{x}_a - \bar{\mathbf{x}}_b \right) \Big) \tag{68b}$$

or

$$\dot{\mathbf{v}}_a = -\sum_{b,c} \left(m_b \left(\partial_\rho \epsilon_a + \partial_\rho \epsilon_b \right) \nabla w_{ab} - \frac{m_b}{m_a} w_{ab} w_{bc} \mathbf{S}_b^{double} \mathbf{x}_{ca} - \frac{m_c}{m_a} w_{cb} w_{ac} \mathbf{S}_c^{double} \mathbf{x}_{ba} \right) \tag{68c}$$

$$\mathbf{A}_a^{double} := \mathbf{P}^{double} \left(\mathbf{Q}^{double} \right)^{-1} = \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{X}_{bc} \otimes \mathbf{x}_{bc} \right) \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{x}_{bc} \otimes \mathbf{x}_{bc} \right)^{-1} \tag{68d}$$

where

$$\mathbf{S}_a^{double} = c_s^2 \left(\mathbf{A}_a^{double}\right)^T \mathbf{A}_a^{double} \mathrm{dev}\left(\left(\mathbf{A}_a^{double}\right)^T \mathbf{A}_a^{double}\right) \left(\mathbf{Q}^{double}\right)_a^{-1}$$
$$\partial_\rho \epsilon_a = c_0^2 \rho_0 \left(\frac{1}{\rho_a^2} + \frac{\rho_0}{\rho_a^3}\right) \tag{68e}$$

It is worth comparing the shear terms for both $\mathbf{A}_a$ and $\mathbf{A}_a^{double}$. The shear force in the first case depends on the pairwise distance $\mathbf{x}_a - \mathbf{x}_b$ and in the second case it depends on the distance of particle a from the center of mass of its neighbours $\mathbf{x}_a - \bar{\mathbf{x}}_b$ which in return yields one higher order convergence rate.

However, this has certain disadvantages as well. If the particle cloud is smaller on one side than on the other eg near the boundaries the formulation becames unstable very quickly. This is a know caveat of similar methods.

This formulation, being derived from energy does conserve energy and conserves linear momentum by the same argument as in previous section.

Numerics

In this section we will test our previous findings. We will first formulate our equations and go over the integration scheme. We will provide a numerical stabilization for tensile instability which is a well know and documented SPH numerical artifiact. Next we will proceed with a theoretical viewpoint answering why a subset of our methods kept failing. Finally we will test our finding on a Beryllium plate benchmark problem. All the numerical code is based on the *SmoothedParticles.jl* Julia library [4, 2, 3].

We are presented with two possible options on how to proceed.

Firstly, we can work directly with $\mathbf{A}_a$ from the the minimalization problem:

$$\mathbf{A}_a := \mathbf{P}_a \mathbf{Q}_a^{-1} = \left(\sum_b w_{ab} \mathbf{X}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1}$$

compute it each time step and get the relation for force, for $\mathbf{A}_a^{double}$, we can proceed equivalently.

Secondly, we can make use of the relation $\mathrm{d}\mathbf{A} = -\mathbf{A}\mathrm{d}\mathbf{L}$ and instead of computing $\mathbf{A}$ directly, we will compute $\mathbf{L}$ each time step and update $\mathbf{A}$ accordingly.

Recall that for $\mathbf{A}_a$ we have:

$$\mathbf{L}_a := \left(\sum_b w_{ab} \mathbf{v}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1}$$

And similarly for $\mathbf{A}_a^{double}$ we have:

$$\mathbf{L}_a^{double} = \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{v}_{bc} \otimes \mathbf{x}_{bc}\right) \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{x}_{bc} \otimes \mathbf{x}_{bc}\right)^{-1}$$

We are than presented with the following system of ODEs:

$$\dot{\mathbf{x}}_a = \mathbf{v}_a \tag{69a}$$

$$m_a \dot{\mathbf{v}}_a = -\mathbf{f}_a \tag{69b}$$

$$\dot{\mathbf{A}}_a = -\mathbf{A}_a \mathbf{L}_a \quad \text{or} \quad (\dot{\mathbf{A}_a^{double}}) = -\mathbf{A}_a^{double} \mathbf{L}_a^{double} \tag{69c}$$

To solve it we proceed as sugested by Kincl [3] utilizing the following combination of midpoint-rule (for $\mathbf{A}$) and Verlet integration scheme (for $\mathbf{x}$ and $\mathbf{v}$):

$$\begin{aligned}
\mathbf{v}_a(t_{k+1/2}) &= \mathbf{v}_a(t_k) + \frac{\Delta t}{2m_a}\mathbf{f}_a(t_k) \\
\mathbf{x}_a(t_k) &= \mathbf{x}_a(t_k) + \frac{\Delta t}{2}\mathbf{v}_a(t_{k+1/2}) \\
\mathbf{A}_a(t_{k+1}) = \mathbf{A}_a(t_k)\left(\mathbf{I} - \frac{\Delta t}{2}\mathbf{L}_a(t_{k+1/2})\right) &\left(\mathbf{I} + \frac{\Delta t}{2}\mathbf{L}_a(t_{k+1/2})\right)^{-1} \\
\mathbf{x}_a(t_{k+1}) &= \mathbf{x}_a(t_{k+1/2}) + \frac{\Delta t}{2}\mathbf{v}_a(t_{k+1/2}) \\
\mathbf{v}_a(t_{k+1}) &= \mathbf{v}_a(t_{k+1/2}) + \frac{\Delta t}{2m_a}\mathbf{f}_a(t_{k+1})
\end{aligned} \tag{70}$$

where $t_k = k\Delta t$ denotes the k-th time step and

$$\mathbf{f}_a(t) = \mathbf{f}_a(\mathbf{x}(t), \mathbf{A}(t)), \quad \mathbf{L}_a = \mathbf{L}_a(\mathbf{x}(t), \mathbf{v}(t))$$

Once again, the simulation suffers from tensile instability - a well-known and documented SPH numerical artifact. To deal with it, we once again follow Kinc [3] and we will add a following repulsive term to the force:

$$\mathbf{f}_a^{penalty} = -\sum_b m_a m_b c_p^2 \frac{\lambda_a + \lambda_b}{\rho_0^2} \frac{h}{r_{ab}} \frac{\partial^2 w_{ab}}{\partial r_{ab} \partial h} \mathbf{x}_{ab}$$

This term is generated by a potential which has the advantage that the (perturbed) total energy still stays constant during the simulation.

Numerical analysis

In this section we want take a more detailed look on the two possible treatments of our system. We will show that one of them is inherently unstable, and we will show consistensty with our equations for the other one.

Recall that our goal is to (numericaly) solve the following ODE system:

$$\dot{\mathbf{x}} = \mathbf{v} \tag{71a}$$

$$m\dot{\mathbf{v}} = \mathbf{f}(\mathbf{A}, \dots) \tag{71b}$$

$$\dot{\mathbf{A}} = -\mathbf{A}\mathbf{L} \tag{71c}$$

Where $\mathbf{x}$ is eulerian position, $\mathbf{X} = \mathbf{x}(t = 0)$ is lagrangian position, $\mathbf{v}$ is velocity, $\mathbf{f}$ is force, m is mass, $\mathbf{A}$ is the distortion field (inverse of deformation gradient $\mathbf{F}$) and $\mathbf{L}$ is the velocity gradient.

We must pay speciall attention to the last equation. First recall that $\mathbf{L}$ is the generator of the one parameter C_0-semigroup $\mathbf{A}(t)$ ie

$$\mathbf{A}(t + \Delta t) \approx \mathbf{A}(t)\exp(-\Delta t\mathbf{L})$$

so while $\mathbf{A}(t)$ lives in $\mathrm{GL}(n)$ and its evolution might be highly non-linear, $\mathbf{L}(t)$ lives in $\mathfrak{gl}(n)$ which is nice vector space.

This is especially important, as this defines the flow (evolution) of the distortion field $\mathbf{A}$ and any numericall approximation must satisfy this property.

To proceed with solving the system we need to approximate either $\mathbf{A}$ or $\mathbf{L}$. We will do so, by approximating the continuum by a system of particles - each particle (labeled b) carries its own internal variables - $\mathbf{x}_b$, $\mathbf{v}_b$, $\mathbf{A}_b$, $\mathbf{L}_b$ etc. The idea is to than evaluate $\mathbf{A}$ at an arbitrary position $\mathbf{x}$ as a projection on this set of particles weighted by w (which can be though of as quadrature weights, or a mollifying kernel). This method is usually referred to as MLS - Moving Least Squares.

There are two possibilities on how to approximate $\dot{\mathbf{A}} = -\mathbf{A}\mathbf{L}$ one is:

$$\mathrm{proj}_{\mathrm{GL}(n)}\mathbf{A}(\mathbf{x}) := \underset{\mathbf{A}\in\mathrm{GL}(n)}{\mathrm{argmin}} \sum_b w(\mathbf{x} - \mathbf{x}_b)\|\mathbf{A}(\mathbf{x} - \mathbf{x}_b) - (\mathbf{X} - \mathbf{X}_b)\|^2 \tag{72}$$

and the other is:

$$\mathrm{proj}_{\mathfrak{gl}(n)}\mathbf{L}(\mathbf{x}) := \underset{\mathbf{L}\in\mathfrak{gl}(n)}{\mathrm{argmin}} \sum_b w(\mathbf{x}-\mathbf{x}_b)\|\mathbf{L}(\mathbf{x}-\mathbf{x}_b)-(\mathbf{v}-\mathbf{v}_b)\|^2 \tag{73}$$

That is to compute the weighted error for each data point (=particle). And the two respective systems of equations are:

$$\mathbf{A}^{n+1} = \mathrm{proj}_{GL}\mathbf{A} \tag{74a}$$

$$\mathbf{x}^{n+1} = \mathbf{x}^n + \Delta t \cdot \mathbf{v}^n \tag{74b}$$

$$\mathbf{v}^{n+1} = \mathbf{v}^n + \Delta t \cdot \mathbf{f}(\mathbf{A}^n, \dots)/m \tag{74c}$$

and:

$$\mathbf{A}^{n+1} = \mathbf{A}^n - \Delta t \cdot \mathbf{A}^n \cdot \mathrm{proj}_{\mathfrak{gl}(n)}\mathbf{L} \tag{75a}$$

$$\mathbf{x}^{n+1} = \mathbf{x}^n + \Delta t \cdot \mathbf{v}^n \tag{75b}$$

$$\mathbf{v}^{n+1} = \mathbf{v}^n + \Delta t \cdot \mathbf{f}(\mathbf{A}^n, \dots)/m \tag{75c}$$

In the first case we are directly projection onto $\mathrm{GL}(n)$ and in the later one onto $\mathfrak{gl}(n)$. Even though both projections are perfectly defined, we have to check if we have have the semigroup property:

$$\mathbf{A}^{n+1} \approx \mathbf{A}^n \exp(-\Delta t \mathbf{L}^n)$$

In the first case this will not be true anymore. Suppose:

$$\mathbf{A}^n = \mathrm{proj}_{\mathrm{GL}(n)}\mathbf{A}(t_n) = \mathbf{A}(t_n) + e_n$$

$$\mathbf{A}^{n+1} = \mathrm{proj}_{\mathrm{GL}(n)}\mathbf{A}(t_{n+1}) = \mathbf{A}(t_{n+1}) + e_{n+1}$$

as there is no link between the residuals e_n and e_{n+1} in general we have:

$$\mathbf{A}^{n+1} - \mathbf{A}^n = \mathbf{A}(t_{n+1}) - \mathbf{A}(t_n) + e_{n+1} - e_n = \mathbf{A}(t_{n+1}) - \mathbf{A}(t_n) + \text{projection noise.}$$

and therefore

$$\mathbf{A}^{n+1} \not\approx \mathbf{A}^n \exp(-\Delta t \mathbf{L}^n)$$

as the projection noise is of order $O(h^p)$ and cannot be refined by smaller timesteping.

This does not mean that

$$\mathrm{proj}_{\mathrm{GL}(n)}\mathbf{A}(t)$$

is a bad approximation of $\mathbf{A}$. It is an approximation that is consistent in space, however the whole system is not consistent in time and it will not give us the correct evolution for $\mathbf{A}$. Our numerical experiments are in agreement with this theoretical finding.

Now we will examine the second case. First assume that the projection $\mathrm{proj}_{\mathfrak{gl}(n)}\mathbf{L}$ approximates $\mathbf{L}$ up to order p in space ie

$$\mathbf{L}_h = \mathrm{proj}_{\mathfrak{gl}(n)}\mathbf{L} = \mathbf{L} + O(h^p)$$

which combined with the implicit euler integration scheme gives:

$$\mathbf{A}^{n+1} = \mathbf{A}^n - \Delta t \cdot \mathbf{A}^n \cdot \mathbf{L}_h^n + O(h^p\Delta) = \mathbf{A}^n\left(\mathbf{I} - \Delta t \mathbf{L}_h^n\right) + O(h^p \Delta t) = \mathbf{A}^n \exp(-\Delta t \mathbf{L}_h^n) + O(\Delta t^2) + O(h^p\Delta t)$$

A similar result will also hold for our mid-point rule
(which is the Cayley map $x \to (1-x/2)(1+x/2)^{-1}$ in disguise):

$$\mathbf{A}^{n+1} = \mathbf{A}^n \left(\mathbf{I} - \frac{\Delta t}{2}\mathbf{L}^{n+1/2}\right)\left(\mathbf{I} + \frac{\Delta t}{2}\mathbf{L}^{n+1/2}\right)^{-1}$$

let $\mathbf{K} = \Delta t \mathbf{L}^{n+1/2}$ than

$$\mathbf{A}^{n+1} = \mathbf{A}^n \left(\mathbf{I} - \frac{1}{2}\mathbf{K}\right)\left(\mathbf{I} + \frac{1}{2}\mathbf{K}\right)^{-1}.$$

expand the inverse assuming ($\Delta t\|\mathbf{L}\| << 1$):

$$\left(\mathbf{I}+\frac{1}{2}\mathbf{K}\right)^{-1}=\mathbf{I}-\frac{1}{2}\mathbf{K}+\frac{1}{4}\mathbf{K}^2-\frac{1}{8}\mathbf{K}^3+\cdots.$$

$$\left(\mathbf{I}-\frac{1}{2}\mathbf{K}\right)\left(\mathbf{I}+\frac{1}{2}\mathbf{K}\right)^{-1}=\mathbf{I}-\mathbf{K}+\frac{1}{2}\mathbf{K}^2-\frac{1}{4}\mathbf{K}^3+O(\|\mathbf{K}^4\|).$$

comparing this with the exponential:

$$\exp(\mathbf{K})=\mathbf{I}-\mathbf{K}+\frac{1}{2}\mathbf{K}^2-\frac{1}{6}\mathbf{K}^3+O(\|\mathbf{K}^4\|).$$

so we have:

$$\mathbf{A}^{n+1}=\mathbf{A}^n\exp(-\Delta t\mathbf{L}^{n+1/2})+O(\Delta t^3).$$

and we can proceed similarly for any other consistent time integrator.

This shows, that even though the idea of Pavelka & Hutter [1] to treat $\mathbf{A}$ as multiple independent MLS projections is appealing in theory, it is sadly not a suitable choice for numerical simulations.

In the next section we will perform numerical experiments validating our findings from previous sections. We will work with the standard Beryllium plate benchmark used in the SHTC community.

Beryllium plate

In this two-dimensional benchmark we prescribe a velocity field to a perfectly elastic beryllium plate and we follow the position of its middle point.

The simulation domain Ω is a symmetric rectangular plate:

$$\Omega=\left(-\frac{L}{2},\frac{L}{2}\right)\times\left(-\frac{W}{2},\frac{W}{2}\right)$$

with $W=0.01$ m and $L=0.06$ m.

The initial velocity field $\mathbf{v}_{init}$ is given as:

$$\mathbf{v}_{init}(\begin{pmatrix}x\\y\end{pmatrix})=A\omega\left(a_1(\sinh s+\sin s)-a_2(\cosh s+\cos s)\right)\begin{pmatrix}0\\1\end{pmatrix}$$

with:

$$s=\alpha\left(x+\frac{L}{2}\right)$$
$$A=4.3369\cdot 10^{-5}\ \mathrm{m}$$
$$\omega=2.3597\cdot 10^{5}\ \mathrm{s}^{-1}$$
$$\alpha=78.834$$
$$a_1=56.6368$$
$$a_2=57.6455$$

Furthermore we assume perfect elasticity $\tau=\infty$ and we set the longitudinal and shear sound speed as:

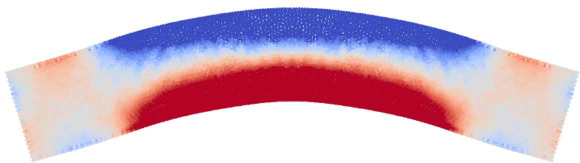
$$c_0=c_s=9046.59\ \mathrm{m\ s}^{-1}$$

The time step for the following figures is chosen as $\mathrm{dr}=W/40$ and the time step is chosen according to the CFL condition as:

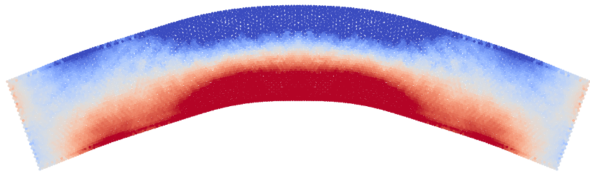
$$\mathrm{dt}=0.05\frac{\mathrm{dr}}{\sqrt{c_0^2+4/3c_s^2}}$$

Add norm(A_evolved - A_recomputed)

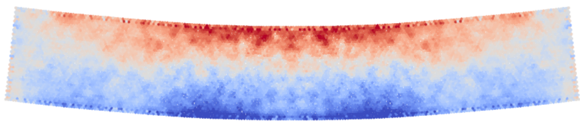
Add figures for A_recomputed failing



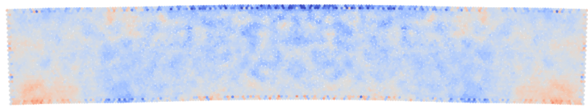
(a) t = 0.075 μs, $c_p/c_0 = 0.04$



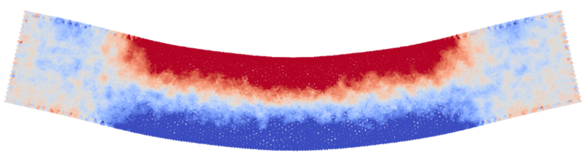
(b) t = 0.075 μs, $c_p/c_0 = 0.04$



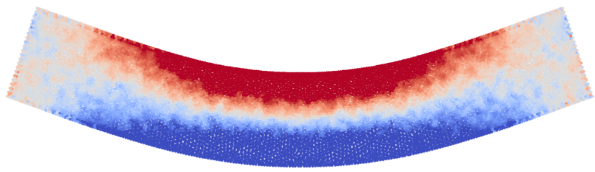
(c) t = 0.150 μs, $c_p/c_0 = 0.04$



(d) t = 0.150 μs, $c_p/c_0 = 0.04$



(e) t = 0.225 μs, $c_p/c_0 = 0.04$



(f) t = 0.225 μs, $c_p/c_0 = 0.04$

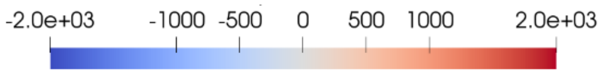
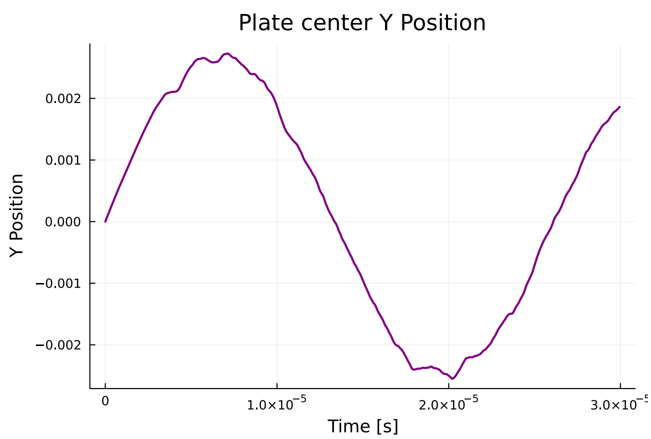


Figure 1: A comparison of the pressure fields for the MLSEvolved and SingleSumEvolved definition of the distortion field in the Beryllium benchmark problem.



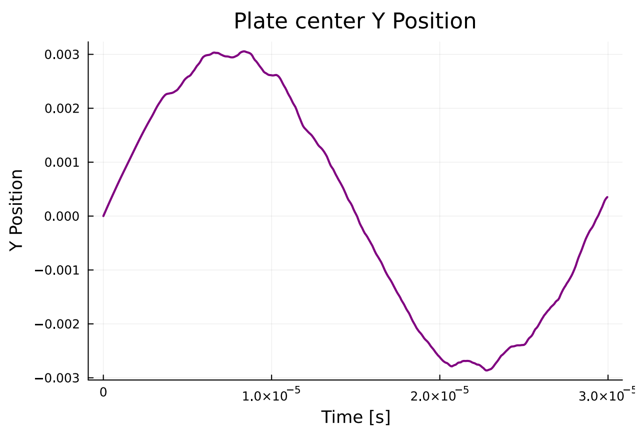


Figure 2: The y coordinate of the center point of the beryllium plate benchmark. (SingleSumEvolved left, MLSEvolved right)

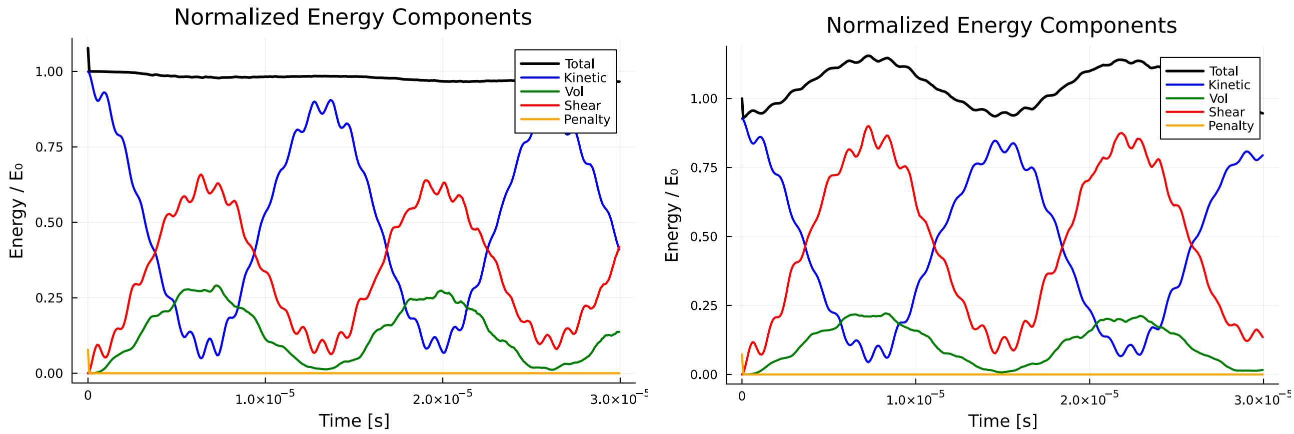


Figure 3: Normalized components of energy plotted against time.
(SingleSumEvolved left, MLSEvolved right)

Conclusions

Write down a proper conclusion

In this paper we have introduced and compared two ideologicaly distinct families of numerical discretization of the SHTC evolution equation for the distortion field 2 withing the SPH framework inspired by [1].

Firstly we had defined the distortion field as a weighted LS projection $\mathrm{proj}_{\mathrm{GL}(n)} : \mathrm{GL}(n) \to \mathrm{GL}(n)$ where we measured the error as $\mathbf{Ax} - \mathbf{X}$. Secondly we had defined the distortion field as a weighted LS projection $\mathrm{proj}_{\mathfrak{gl}(n)} : \mathrm{GL}(n) \to \mathfrak{gl}(n)$ where we measured the error as $\mathbf{Lx} - \mathbf{v}$.

We have shown that the first projection $\mathbf{A} \to \mathrm{proj}_{\mathrm{GL}(n)}\mathbf{A}$ while an excellent approximation of $\mathbf{A}$ at a fixed time will not yield the correct dynamics while the second projection $\mathbf{L} \to \mathrm{proj}_{\mathrm{GL}(n)}\mathbf{L}$ will yield the correct dynamics for $\mathbf{A}$.

We have suggested several discretizations (summed up in the appendix) related by $\mathrm{d}\left(\mathrm{proj}_{\mathrm{GL}(n)}\mathbf{A}\right)(\mathbf{x}) = -\left(\mathrm{proj}_{\mathrm{GL}(n)}\mathbf{A}\right)(\mathbf{x})\left(\mathrm{proj}_{\mathfrak{gl}(n)}\mathbf{L}\right)(\mathbf{x}) + O(\left(\mathrm{proj}_{\mathrm{GL}(n)}\mathbf{A}\right)\mathbf{x} - \mathbf{X})$ with the error $O(\left(\mathrm{proj}_{\mathrm{GL}(n)}\mathbf{A}\right)\mathbf{x} - \mathbf{X})$ being the smallest possible. Next we have derived the equations of motion for both formulations and proved several usefull properties regarding conservation of energy, linear and angular momentum, and order of convergence in space and in time.

A discussion putting all the methods in place and unifiying them as a weighted LS problem with various lengths was done. We have arrived at the conlusion that the MLS formulation is the 'correct' formulation for our problem as it is a discrete version of 59 and therefore it does take into account the dynamical properties of individual particles (ie the volume, density and mass) unline the *SingleSum* formulation which does take into account only the geometry (positions). To put it plainly MLS uses the correct quadrature weights which do recover the distortion field in the limit.

We have performed multiple numerical simulations to confirm our findings.

The code used for the simulations is made public and can be found on this adress: https://github.com/MichaelM
distortion-shtc

Add references

Add citations

References

[1] M. Hütter and M. Pavelka. Particle-based approach to the eulerian distortion field and its dynamics. *Continuum Mechanics and Thermodynamics*, 35(5):1943– 1967, 2023.

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[4] O. Kincl, D. Schmoranzer, and M. Pavelka. Simulation of superfluid fountain effect using smoothed particle hydrodynamics. *Physics of Fluids*, 35(047124), 2023.

Definition	Energy functional	Distortion
Kincl	$D[\mathbf{L}](\mathbf{x}_a) = \sum_b m_b w'_{ab}/r_{ab} \lVert \mathbf{L}\mathbf{x}_{ab} - \mathbf{v}_{ab} \rVert^2$	$\mathbf{A}_a = \left(\sum_b m_b w'_{ab}/r_{ab} \mathbf{X}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b m_b w'_{ab}/r_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1}$
MLSEvolved	$D[\mathbf{L}](\mathbf{x}_a) = \sum_b V_b w_{ab} \lVert \mathbf{L}\mathbf{x}_{ab} - \mathbf{v}_{ab} \rVert^2$	$\mathbf{A}_a = \left(\sum_b V_b w_{ab} \mathbf{X}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b V_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1}$
MLS	$D[\mathbf{A}](\mathbf{x}_a) = \sum_b V_b w_{ab} \lVert \mathbf{A}\mathbf{x}_{ab} - \mathbf{X}_{ab} \rVert^2$	$\mathbf{A}_a = \left(\sum_b V_b w_{ab} \mathbf{X}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b V_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1}$
SingleSumEvolved	$D[\mathbf{L}](\mathbf{x}_a) = \sum_b w_{ab} \lVert \mathbf{L}\mathbf{x}_{ab} - \mathbf{v}_{ab} \rVert^2$	$\mathbf{A}_a = \left(\sum_b w_{ab} \mathbf{X}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1}$
SingleSum	$D[\mathbf{A}](\mathbf{x}_a) = \sum_b w_{ab} \lVert \mathbf{A}\mathbf{x}_{ab} - \mathbf{X}_{ab} \rVert^2$	$\mathbf{A}_a = \left(\sum_b w_{ab} \mathbf{X}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1}$
DoubleSumEvolved	$D[\mathbf{L}](\mathbf{x}_a) = \sum_{b,c} w_{ab} w_{ac} \lVert \mathbf{L}\mathbf{x}_{bc} - \mathbf{v}_{bc} \rVert^2$	$\mathbf{A}_a = \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{X}_{bc} \otimes \mathbf{x}_{bc}\right) \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{x}_{bc} \otimes \mathbf{x}_{bc}\right)^{-1}$
DoubleSum	$D[\mathbf{A}](\mathbf{x}_a) = \sum_{b,c} w_{ab} w_{ac} \lVert \mathbf{A}\mathbf{x}_{bc} - \mathbf{X}_{bc} \rVert^2$	$\mathbf{A}_a = \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{X}_{bc} \otimes \mathbf{x}_{bc}\right) \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{x}_{bc} \otimes \mathbf{x}_{bc}\right)^{-1}$

Definition	Velocity gradient	Projection space and Projected variable
Kincl	$\mathbf{L}_a = \left(\sum_b m_b w'_{ab}/r_{ab} \mathbf{v}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b m_b w'_{ab}/r_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1}$	$\mathfrak{gl}(n)$ and $\mathbf{L}$
MLSEvolved	$\mathbf{L}_a = \left(\sum_b V_b w_{ab} \mathbf{v}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b V_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1}$	$\mathfrak{gl}(n)$ and $\mathbf{L}$
MLS	$\mathbf{L}_a = \left(\sum_b V_b w_{ab} \mathbf{v}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b V_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1}$	$\mathrm{GL}(n)$ and $\mathbf{A}$
SingleSumEvolved	$\mathbf{L}_a = \left(\sum_b w_{ab} \mathbf{v}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1}$	$\mathfrak{gl}(n)$ and $\mathbf{L}$
SingleSum	$\mathbf{L}_a = \left(\sum_b w_{ab} \mathbf{v}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1}$	$\mathrm{GL}(n)$ and $\mathbf{A}$
DoubleSumEvolved	$\mathbf{L}_a = \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{v}_{bc} \otimes \mathbf{x}_{bc}\right) \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{x}_{bc} \otimes \mathbf{x}_{bc}\right)^{-1}$	$\mathfrak{gl}(n)$ and $\mathbf{L}$
DoubleSum	$\mathbf{L}_a = \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{v}_{bc} \otimes \mathbf{x}_{bc}\right) \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{x}_{bc} \otimes \mathbf{x}_{bc}\right)^{-1}$	$\mathrm{GL}(n)$ and $\mathbf{A}$

Table 1: Comparison of different definitions of the distortion field:
(top) energy functional and resulting distortion; (bottom) velocity gradient and evolved quantity.